Discrete Mathematics
Practice Questions: Algorithms and Integers

Instructions: Students must show all reasoning clearly.

1. Chapter 3.1 Algorithms

1.1. List all the steps used to search for 9 in the sequence 1, 3, 4, 5, 6, 8, 9, 11 using a) a linear search. b) a binary search.

1.2. A palindrome is a string that reads the same forward and backward. Describe an algorithm for determining whether a string of n characters is a palindrome.

1.3. Devise an algorithm that finds the first term of a sequence of positive integers that is less than the immediately preceding term of the sequence.

1.4. Devise an algorithm that finds the first term of a sequence of integers that equals some previous term in the sequence.

1.5. Devise an algorithm that finds a mode in a list of nondecreasing integers.

1.6. The ternary search algorithm locates an element in a list of increasing integers by successively splitting the list into three sublists of equal (or as close to equal as possible) size, and restricting the search to the appropriate piece. Specify the steps of this algorithm.

1.7. Describe an algorithm that will count the number of 1s in a bit string by examining each bit of the string to determine whether it is a 1 bit.

1.8. Describe an algorithm that locates the first occurrence of the largest element in a finite list of integers, where the integers in the list are not necessarily distinct.

2. Chapter 3.2 Growth of Functions

2.1. Find the least integer n such that $f(x)$ is $O(x^n)$ for each of these functions.

- $f(x) = 2x^3 + x^2 \log x$
- $f(x) = 3x^3 + (\log x)^4$
- $f(x) = (x^4 + x^2 + 1)/(x^3 + 1)$
- $f(x) = (x^4 + 5 \log x)/(x^4 + 1)$

2.2. Show that 2^n is $O(3^n)$ but that $3n$ is not $O(2n)$.

2.3. Arrange the functions $\sqrt{n}, 1000\log n, n\log n, 2n!, 2n, 3n,$ and $n^2/1,000,000$ in a list so that each function is big-O of the next function.

2.4. Give as good a big-O estimate as possible for each of these functions.

- $(n^2+8)(n+1)$
- $(n\log n + n^2)(n^3+2)$
- $(n! + 2n)(n^3 + \log(n^2+1))$

3. Chapter 3.3 Complexity of Algorithms

3.1. Give a big-O estimate for the number of operations (where an operation is an addition or a multiplication) used in this segment of an algorithm.

```
t := 0
for i := 1 to 3 do
  for j := 1 to 4 do
    t := t + i*j
```

3.2. Give a Big-O estimate for the number of operations, where an operation is defined as either a comparison or a multiplication, used in the following segment of an algorithm (ignoring comparisons used to test the conditions in the for loops, where $a_1, a_2, \ldots, a_n$ are positive real numbers):

```
m := 0
for i := 1 to n do
  for j := i+1 to n do
    m := max(a_i a_j, m)
```

3.3. Describe how the number of comparisons used in the worst case changes when these algorithms are used to search for an element of a list when the size of the list doubles from n to 2n, where n is a positive integer. a) linear search b) binary search

4. Chapter 4.1 Divisibility and Modular Arithmetic

4.1. Find values $(-133 \mod 23 + 261 \mod 23) \mod 23$, and $(457 \mod 23 \times 182 \mod 23) \mod 23$.

4.2. Find each of these values.

$$(992 \mod 32)^3 \mod 15$$

$$(34 \mod 17)^2 \mod 11$$

$$(193 \mod 23)^2 \mod 31$$

$$(893 \mod 79)^4 \mod 26$$

5. Chapter 4.2 Integer Representations and Algorithms

5.1. Convert the hexadecimal expansion of each of these integers to a binary expansion. $(80E)_{16}$, $(135AB)_{16}$, $(ABBA)_{16}$, $(DEFACED)_{16}$.

5.2. Convert $(101101111011)_2$ from its binary expansion to its hexadecimal expansion.

5.3. Find the sum and the product of each of these pairs of numbers. Express your answers as a binary expansion.

$$(1000111)_2, (1110111)_2$$

$$(11101111)_2, (10111101)_2$$

$$(1010101010)_2, (111110000)_2$$

$$(1000000001)_2, (1111111111)_2$$

5.4. Find $7^{644} \mod 645$, $3^{2003} \mod 99$.

6. Chapter 4.3 Primes and Greatest Common Divisors

7.1. Find $\gcd(92928, 123552)$and $\operatorname{lcm}(92928, 123552)$, and verify that $\gcd(92928, 123552) \times \operatorname{lcm}(92928, 123552) = 92928 \times 123552$

7.2. Use the Euclidean algorithm to find

$$\gcd(12, 18)$$

$$\gcd(111, 201)$$

$$\gcd(1001, 1331)$$

$$\gcd(12345, 54321)$$

$$\gcd(1000, 5040)$$

$$\gcd(9888, 6060)$$

7. Chapter 6 Counting

6.1. How many ways are there to choose 6 items from 10 distinct items when

- the items in the choices are ordered and repetition is not allowed?
- the items in the choices are ordered and repetition is allowed?
- the items in the choices are unordered and repetition is not allowed?
- the items in the choices are unordered and repetition is allowed?

6.2. An ice cream parlor has 28 different flavors, 8 different kinds of sauce, and 12 toppings.

- In how many different ways can a dish of three scoops of ice cream be made where each flavor can be used more than once and the order of the scoops does not matter?
- How many different kinds of small sundaes are there if a small sundae contains one scoop of ice cream, a sauce, and a topping?
- How many different kinds of large sundaes are there if a large sundae contains three scoops of ice cream, where each flavor can be used more than once and the order of the scoops does not matter; two kinds of sauce, where each sauce can be used only once and the order of the sauces does not matter; and three toppings, where each topping can be used only once and the order of the toppings does not matter?

6.3. Show that in a sequence of m integers there exists one or more consecutive terms with a sum divisible by m.

6.4. Show that the decimal expansion of a rational number must repeat itself from some point onward.

6.5. Find n for each of the following statements $C(n, 2) = 45$. $C(n, 3) = P(n, 2)$. $C(n, 5) = C(n, 2)$.

6.6. Suppose that S is a set with n elements. How many ordered pairs (A, B) are there such that A and B are subsets of S with $A \subseteq B$?

6.7. Let n and r be integers with $1 \leq r < n$. Show that

$$C(n, r - 1) = C(n + 2, r + 1) - 2C(n + 1, r + 1) + C(n, r + 1).$$

6.8. How many bit strings of length n, where $n \geq 4$, contain exactly two occurrences of 01?

6.9. Suppose that when an enzyme that breaks RNA chains after each G link is applied to a 12-link chain, the fragments obtained are $G, CCG, AAAG,$ and $UCCG$, and when an enzyme that breaks RNA chains after each C or U link is applied, the fragments obtained are $C, C, C, C, GGU,$ and $GAAAG$. Can you determine the entire 12-link RNA chain from these two sets of fragments? If so, what is this RNA chain?

6.10. Find the coefficient of $x^3y^2z^5$ in $(x + y + z)^{10}$.

6.11. How many ways are there to travel in xyz space from the origin (0, 0, 0) to the point (3, 2, 5) by taking steps one unit in the positive x direction, one unit in the positive y direction, or one unit in the positive z direction? (Moving in the negative x, y, or z direction is prohibited, so that no backtracking is allowed.)